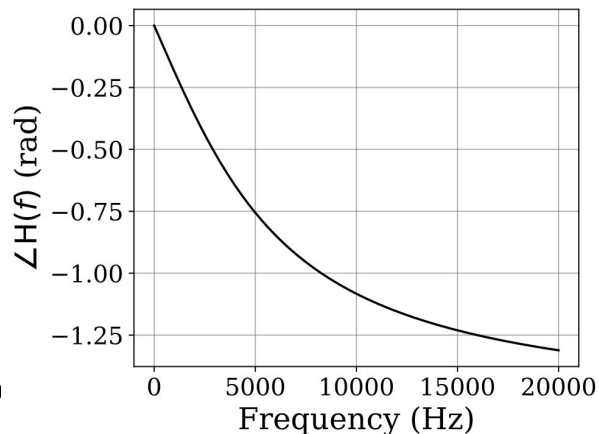
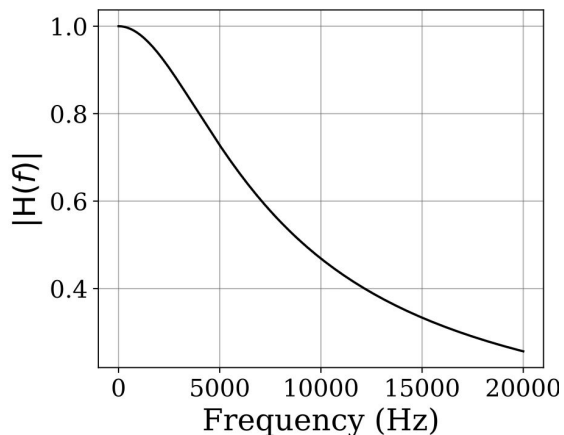

PSDs and Noise

Silvia Zhang
SAGI 2025

Transfer Functions and PSDs

You've been shown for an RC circuit that the 'transfer function' is:

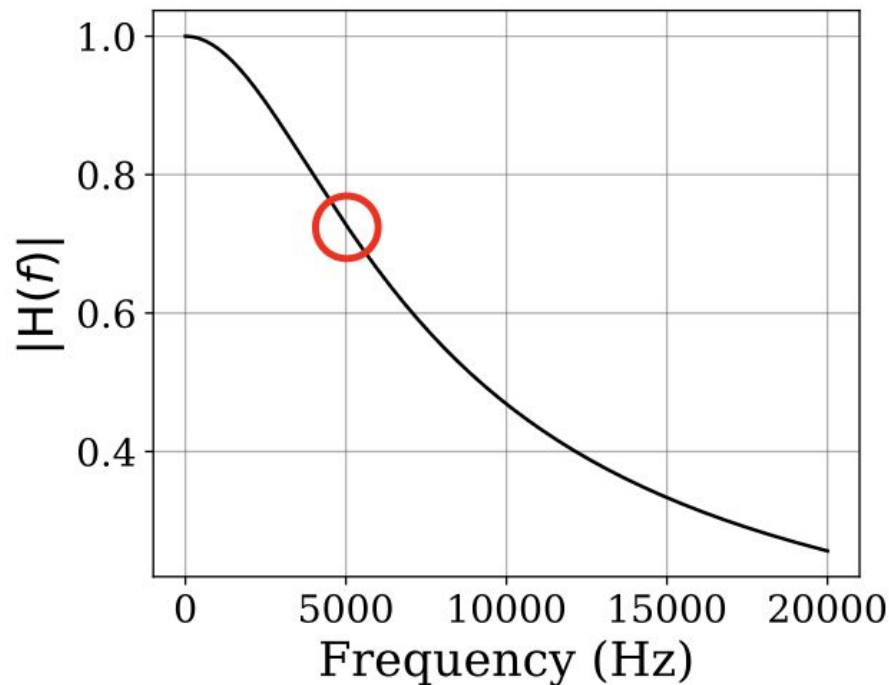
$$|H(\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}, \quad \angle H(\omega) = \tan^{-1}(-\omega RC)$$



Low Pass Filters and Transfer Functions

This is saying that for a 5.3kHz sinusoid, its amplitude will be reduced to $1/\sqrt{2}$.

We call this the **-3db knee** of a filter. These are also used to describe amplifiers.

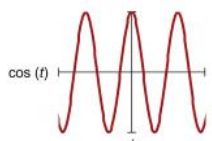
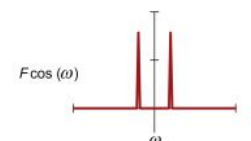
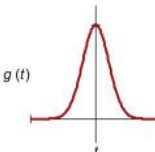
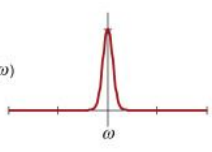
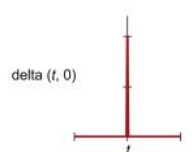
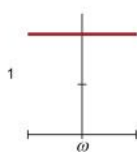
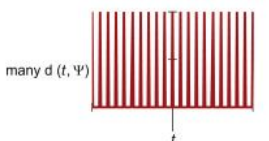
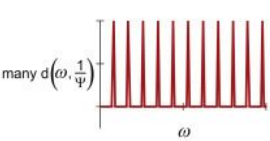


Fourier Transforms

We can represent periodic waveforms with their Fourier coefficients without loss of information!

When looking at signals, try to guess their FT.

$$x_p(t) = \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_0 t} \quad c_n = \frac{1}{T_0} \int_{T_0} x_p(t) e^{-j2\pi n f_0 t} dt$$

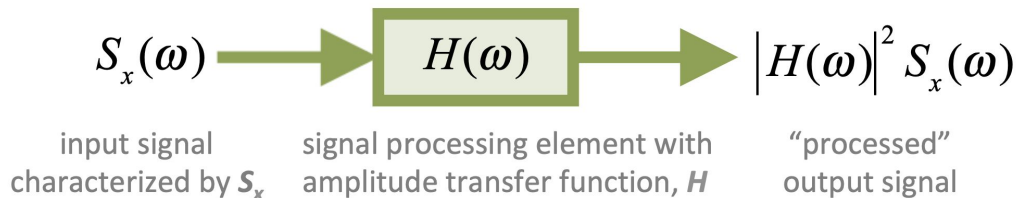
Time domain signals	Frequency domain spectra
 (A) Cosine wave	 (B) Fourier transform of cosine wave
 (C) Gaussian function	 (D) Spectrum of Gaussian function
 (E) Delta function	 (F) Frequency content of delta function
 (G) Sampling function in time domain	 (H) Transform of sampling function

Power Spectral Density (PSD)

$S_x(\omega)$ is the distribution of the energy of a signal over frequency.

$$S_x(\omega) = |X(\omega)|^2$$

Spectral density allows us to characterize signal and noise power over all frequencies. Can allow us to analyze signal processing in frequency domain!

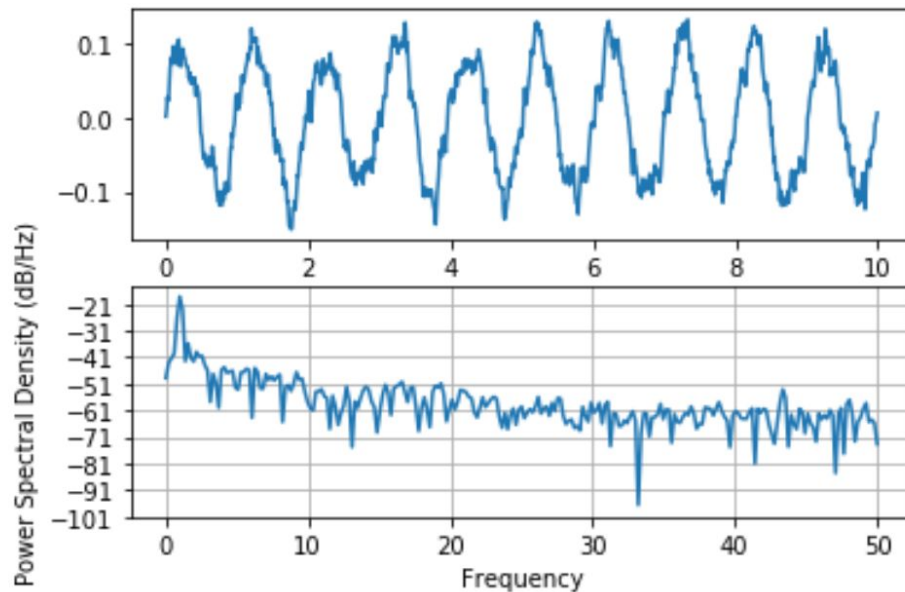


Power Spectral Density (PSD)

An example time stream and its PSD.

Note - if the signal strength was weaker, how would the PSD look?

If we wanted to see this sinusoid more clearly, what would we do?

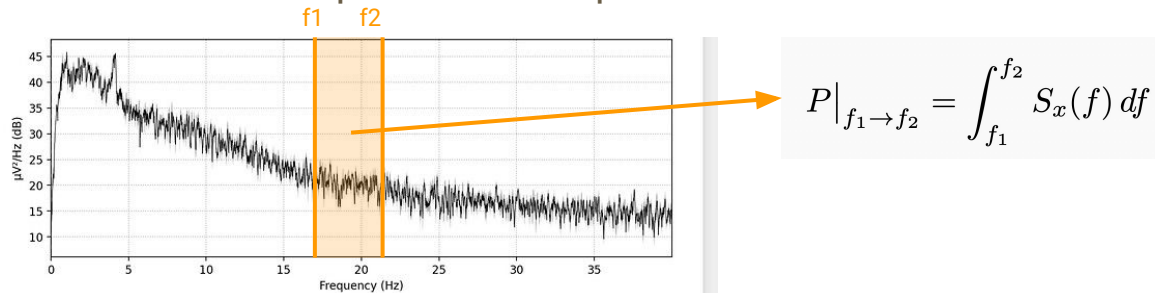


Parseval's Theorem

States that signal power is equivalent in both time and frequency domains:

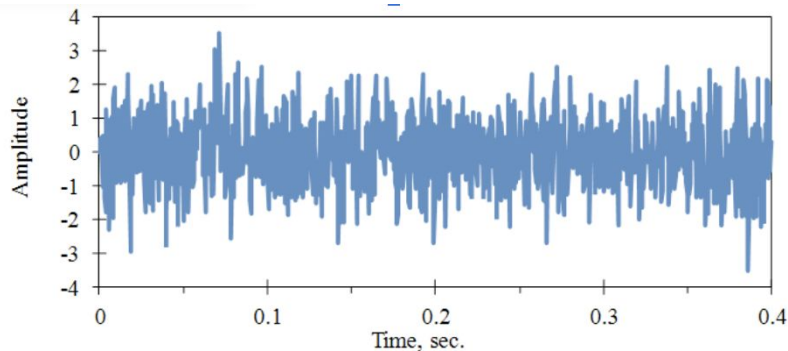
$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt = \int_{-\infty}^{\infty} S_x(f) df$$

This allows us to connect fractional power to a specific bandwidth:



RMS Noise

Rms stands for 'root-mean-square'. Noise usually averages to 0, so we take:



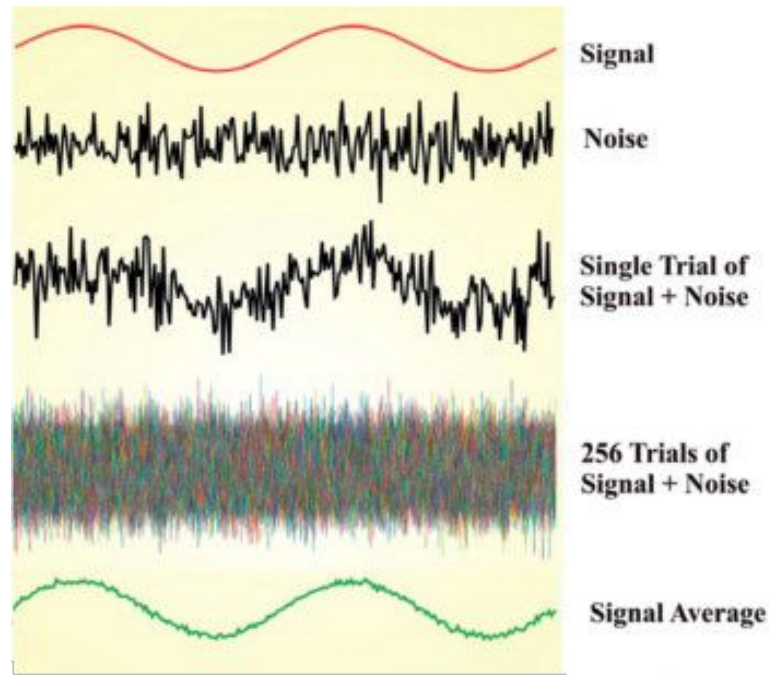
$$x_{\text{RMS}} = \sqrt{\frac{1}{n} (x_1^2 + x_2^2 + \cdots + x_n^2)} \quad \text{var}(N) \equiv N_{\text{rms}}^2 \equiv \langle N(t)^2 \rangle$$

RMS Noise

RMS noise averages down over n samples!

That means for uncorrelated, random noise we can **reduce our noise** by either averaging samples, or taking longer time-streams.

$$x_{\text{RMS}} = \sqrt{\frac{1}{n} (x_1^2 + x_2^2 + \dots + x_n^2)}$$



PSDs vs RMS Quantities

When we start thinking about noise, we can think of it in frequency (PSD) or rms (time) domains.

True if PSD is flat/white!

Voltage noise

$$V_n = \left\{ \frac{1}{2\pi H_{\max}^2} \int_0^\infty d\omega S_{V_n}(\omega) |H(\omega)|^2 \right\}^{1/2} = \sqrt{S_{V_n} \Delta f}$$

PSDs

Rms quantities

Current noise

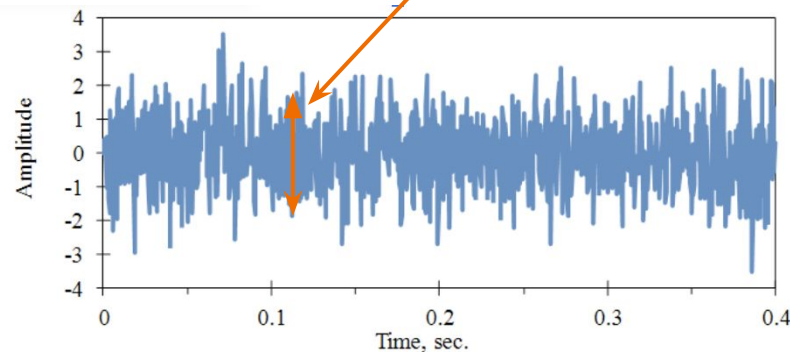
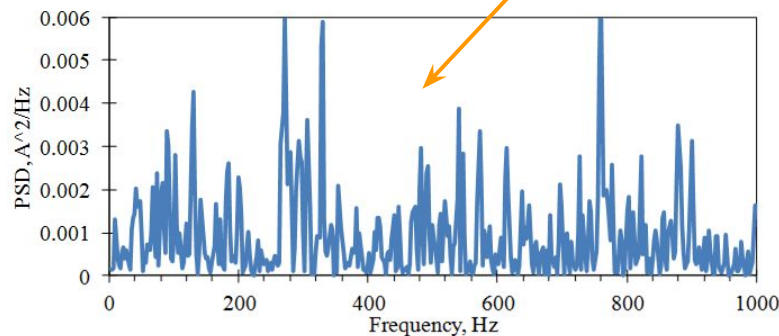
$$I_n = \left\{ \frac{1}{2\pi H_{\max}^2} \int_0^\infty d\omega S_{I_n}(\omega) |H(\omega)|^2 \right\}^{1/2} = \sqrt{S_{I_n} \Delta f}$$

PSDs vs RMS Quantities

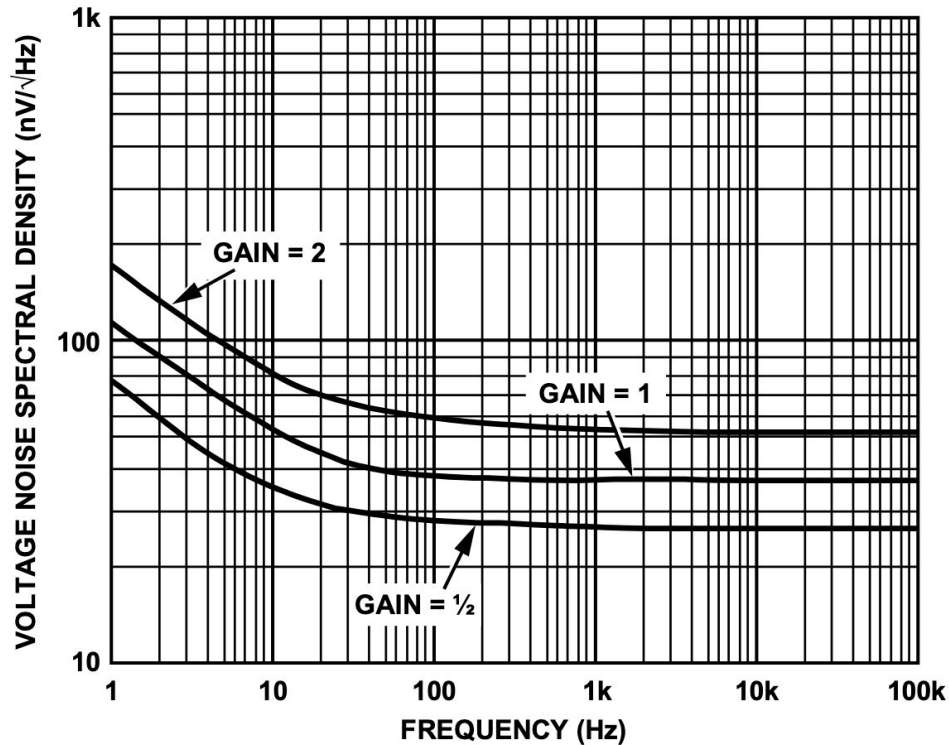
PSDs

$$V_n = \left\{ \frac{1}{2\pi H_{\max}^2} \int_0^\infty d\omega S_{V_n}(\omega) |H(\omega)|^2 \right\}^{1/2} = \sqrt{S_{V_n} \Delta f}$$

Rms quantities



Example: JFET Amplifier Voltage Noise



Noise and Auto Correlation

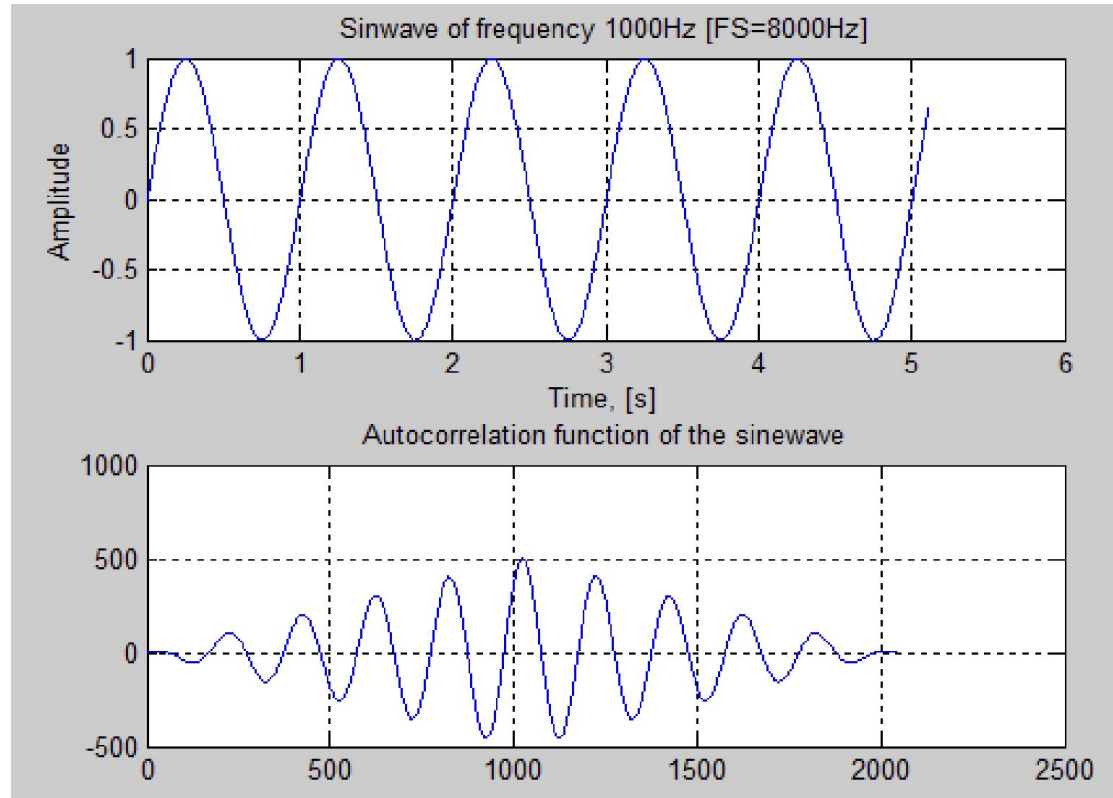
The variances $\text{var}(N) = N \text{rms}^2$ quantifies the noise strength, but doesn't tell us about its time dependence or frequency information.

This is contained in the autocorrelation function:

$$R_{xx}(\tau) = \int_{-\infty}^{\infty} x(t) x(t + \tau) dt$$

This function is peaked when a function is self-similar.

Noise and Auto Correlation



Wiener-Khintchine Theory

Very simply, we can relate the PSD of a stationary, random process (noise) to the Fourier transform of its autocorrelation function.

$$S_{xx}(f) = \int_{-\infty}^{\infty} R_{xx}(\tau) e^{-j2\pi f\tau} d\tau$$



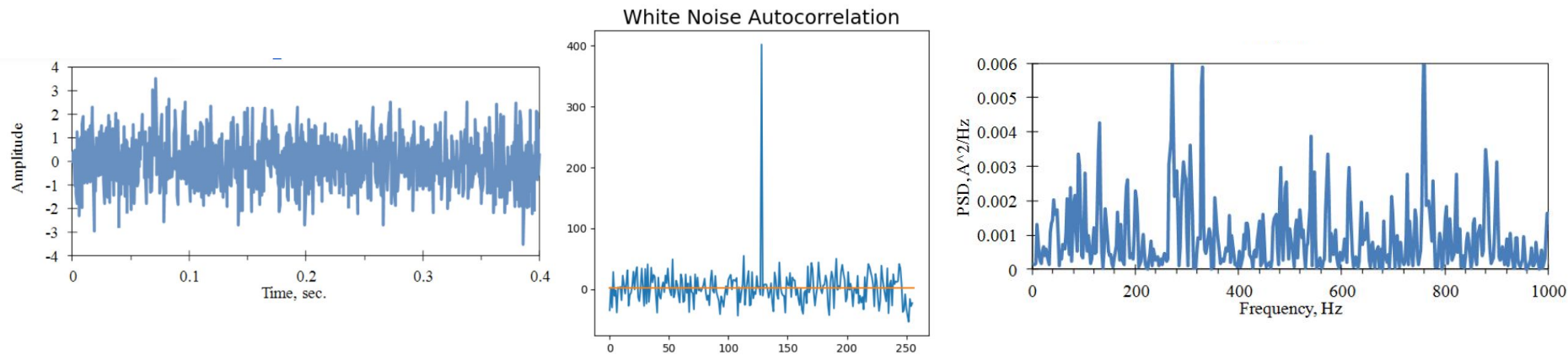
Aleksandr Khinchin



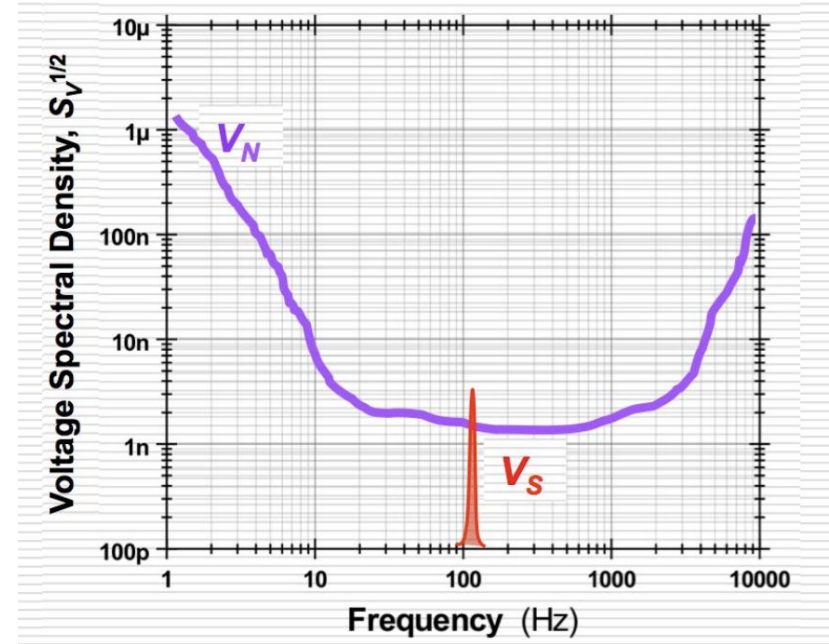
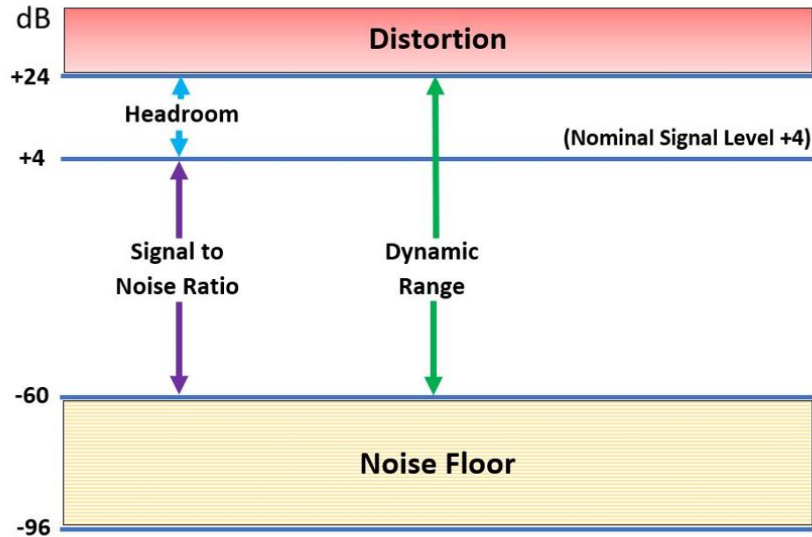
Norbert Wiener 15

Example: Wiener-Khintchine Theory

Think of white noise - this function will only line up with itself at 1τ .



Signal above Noise Floor



Types of Noise

- Thermal Noise (Nyquist/Johnson)
 - Equilibrium thermodynamic fluctuations
- Shot Noise
 - Comes from discrete nature of charge (photons)
- 1/f Noise (Flicker)
 - Originating in
- Quantum
 - TLS noise, zero-point noise (not discussing).

White Noise

'White noise' fundamentally is coming from thermal fluctuations in materials via 'fluctuation dissipation theorem'. **When there is a process that dissipates energy into heat, a reverse process is produced that causes thermal fluctuations.**

White noise is constant at all frequencies within our band.

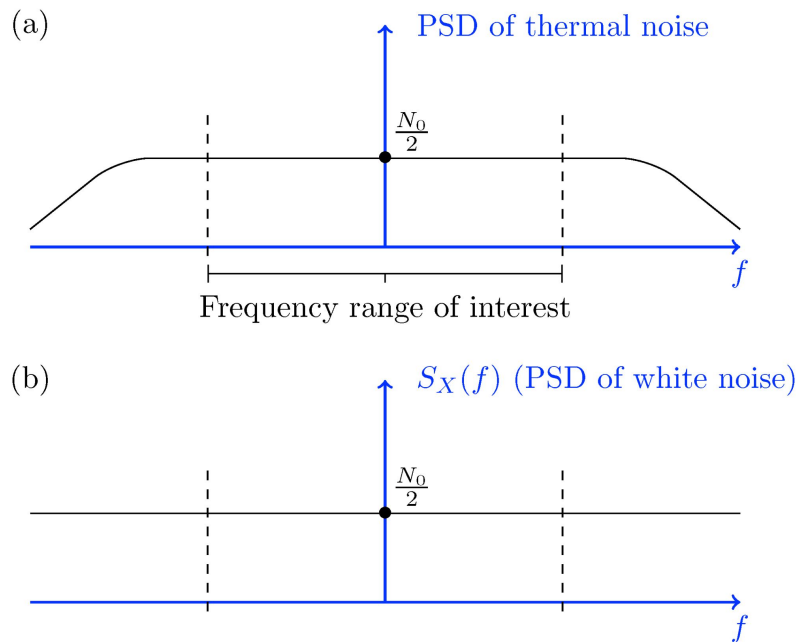
- Drag -> Brownian motion
- Resistance -> Johnson noise
- Light absorption -> thermal radiation

Nyquist/Thermal Noise

- Johnson noise arises in thermal fluctuations of population in electron states
- PSD is white/flat, up to high f .
- High f cutoff, occupation of electron states in material maximized

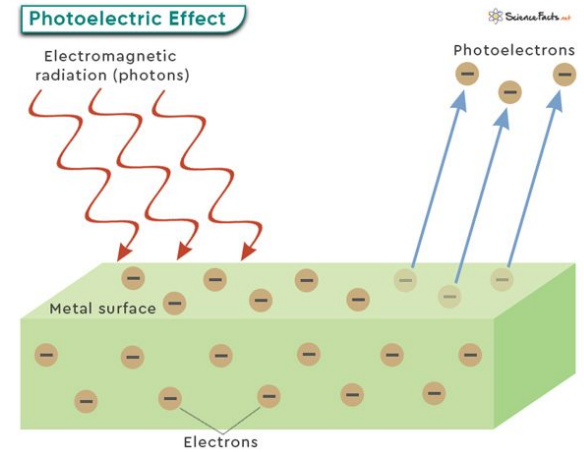
$$V_{\text{rms}} = \sqrt{V_n^2} = \sqrt{4k_B T R \Delta f}$$

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{R} = \sqrt{\frac{4k_B T \Delta f}{R}}$$



Shot Noise

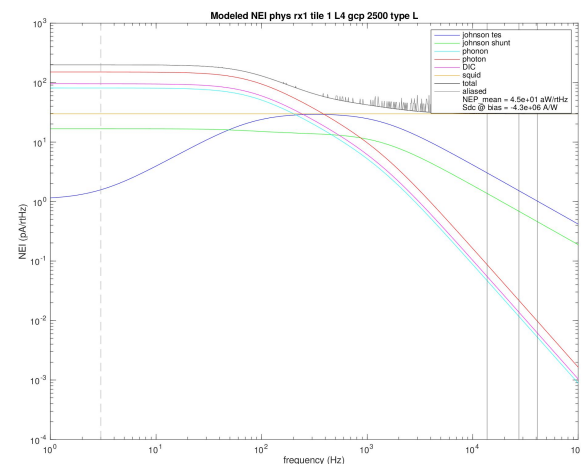
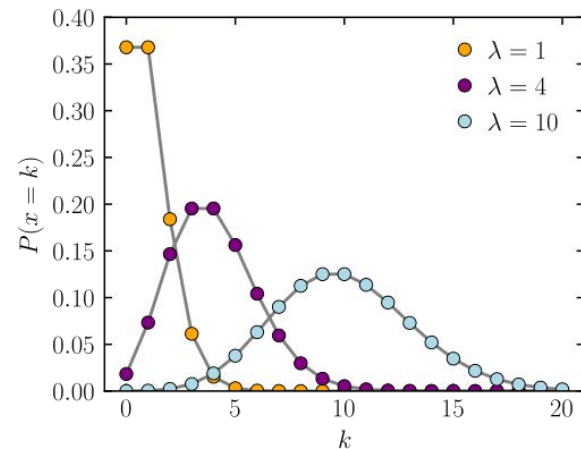
- Charge flows in discrete units (ie.electrons)
- Arrival times of particles is a **random process**.
- Think of rain on a roof - will the amount of water running off be constant? Or will it increase/decrease with the randomness of when raindrops hit the roof?
- Shot noise is a type of **current noise**.



Shot Noise

- Shot Noise is also called 'Poisson Noise', the shape is dependent on the rate event λ . K is # events.
- Shot Noise/photon noise is the dominant noise source in TES detectors, we therefore call them 'quantum-noise limited', we cannot improve the noise performance past the fundamental limit from the quantization of charge.

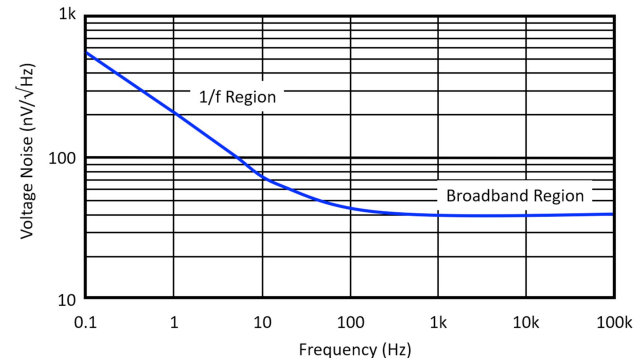
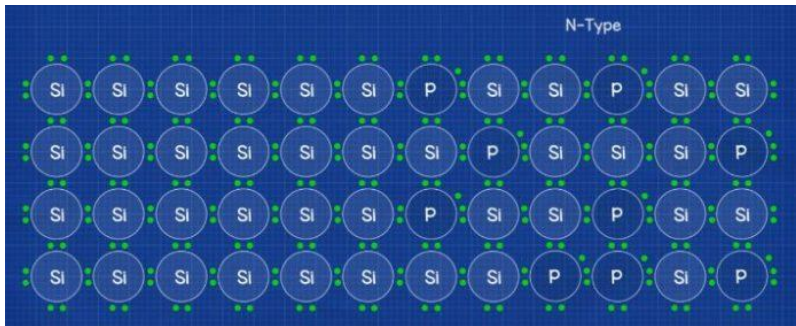
$$S_i(f) = 2qi_0$$



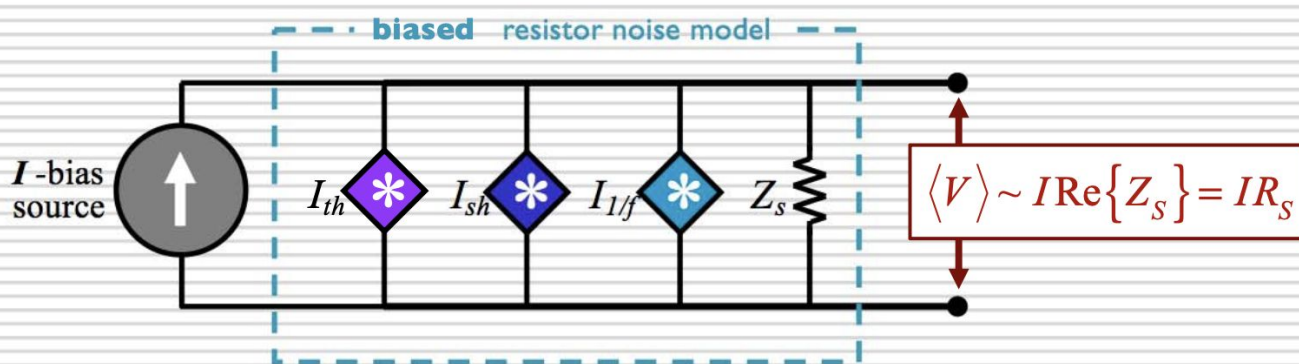
1/f Noise (Flicker Noise)

$$S_{I_{1/f}} = (IR_s)^2 S_g$$

- S_g = PSD of conductance fluctuations.
- Arises in conductance/resistance fluctuations within a material.
- Think semiconductors - randomly doped with other elements. Disorder in conductor introduces fluctuations
- Randomness exists in all materials -> **1/f noise is in all systems.**



Noise Model of a Conductor Under Voltage Bias



Noise Components (ignoring noise from source)

thermal noise: $S_{I_{th}} = 4k_B T^* / \operatorname{Re}\{Z_s\}$

shot noise: $S_{I_{sh}} = \zeta 2eI$, $\zeta =$ “correlation factor”

1/f noise: $S_{I_{1/f}} = (I R_s)^2 S_G = \langle V \rangle^2 S_G$

Example: Noise on BICEP Array

Here we show noise on all detectors in BICEP Array. What types of noise do you see demonstrated here? What are they caused by?

Why do think there's a big drop off at higher frequency?

